

# Joshua Oti

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## EDUCATION

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| <b>University of Southern California</b><br>Master of Science, Computer Science; <b>GPA: 3.76/4.0</b><br>• <b>Relevant Courses:</b> Operating Systems, Networks, Database Systems, Software Engineering, Algorithms                                                                                                           | <b>Los Angeles, CA</b><br>May 2026 |
| <b>Covenant University</b><br>Bachelor of Engineering, Mechanical Engineering; <b>GPA: 4.85/5.0</b><br>• <b>Relevant Courses:</b> Control Systems, Linear Algebra, Modeling & Simulation, Numerical Methods<br>• <b>Academic Honors:</b> Top 1% of Graduating Class (Mechanical Engineering); 3rd Place, ACPC (Regional ICPC) | <b>Ogun, Nigeria</b><br>Oct 2021   |

## TECHNICAL SKILLS

- **Languages:** Python, C, C++, TypeScript, SQL, Java, C#, bash
- **Web/Backend:** FastAPI, REST APIs, React, gRPC, Node.js, Spring Boot, ASP.NET Core, Django, OAuth 2.0, J2EE
- **AI/ML:** PyTorch, MCP, LangChain, RAGAS, LoRA, Transformers, Scikit-learn, ChromaDB, Google AI Studio
- **Tools:** Git, RabbitMQ, Kafka, MLflow, Figma, GraphQL, Grafana, Prometheus, Postman, JMeter
- **Cloud/Infra:** Amazon Web Services, Kubernetes, Microsoft Azure, Google Cloud, Docker, CI/CD
- **Certifications:** AWS Certified Solutions Architect Associate (AWS CSAA)
- **Affiliations:** National Society of Black Engineers (NSBE), CodePath, Association for Computing Machinery

## EXPERIENCE

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| <b>Autodesk</b><br>Machine Learning Engineer Intern<br>• Owned the end-to-end lifecycle of an enterprise GenAI content platform using a FastAPI backend, Azure OpenAI, and AWS S3, automating workflows to save 10+ hours weekly.<br>• Designed dynamic LLM pipelines incorporating few-shot prompting strategies to significantly optimize generative model reasoning and image output quality.<br>• Engineered a responsive Vue.js gamified frontend with enterprise SSO integration, driving a 20% increase in user participation through seamless, multi-device accessibility.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>San Francisco, CA</b><br>May 2025 - Aug 2025 |
| <b>Phasecurve Technologies Ltd</b><br>Software Engineer<br>• Architected a cloud-based ETL and analytics backend using Azure SQL and Flask microservices on Azure App Services, implementing autoscaling to absorb sudden 3x traffic spikes on a fintech platform.<br>• Engineered event-driven data pipelines using Azure Serverless Functions and Azure Queues for asynchronous social media data extraction, implementing exponential backoff to reliably handle strict third-party API rate limits.<br>• Automated CI/CD pipelines leveraging GitHub Actions and Docker, containerizing services to ensure reliable deployments and cutting deployment time by 50%.<br>• Optimized content uploads with multi-threading and asynchronous processing, reducing backend response latency by 80%, and integrated Azure Application Insights for telemetry monitoring to rapidly diagnose pipeline failures.<br>• Led the migration from Mailgun API to Gmail API, optimizing the email routing service and significantly reducing operational costs. | <b>Lagos, Nigeria</b><br>Nov 2022 - Sep 2024    |
| <b>ReHboX</b><br>Machine Learning Engineer<br>• Developed and deployed a real-time computer vision algorithm for edge devices to analyze human pose from camera inputs utilizing Python and OpenCV.<br>• Built a predictive machine learning model using Scikit-learn alongside an interactive data dashboard, providing actionable insights on the Total Addressable Market (TAM).<br>• Coauthored 6 technical design documents, aligning engineering workflows with user stories to streamline development sprints and cut meeting times by 15%.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Lagos, Nigeria</b><br>Jun 2021 - Feb 2022    |

## SELECTED PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |
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| <b>CoursePilot   Python, React, Claude, Firebase — Full-Stack &amp; GenAI</b><br>• Architected and deployed a full-stack AI productivity platform featuring a context-aware RAG Chat Assistant and generative study tools to automate academic workflows.<br>• Developed an automated ingestion pipeline utilizing Claude Sonnet and OCR to parse unstructured syllabi and handwritten notes into structured JSON, integrating a "human-in-the-loop" review interface to ensure 100% calendar event accuracy.<br>• Engineered a Retrieval-Augmented Generation (RAG) AI Assistant capable of executing natural language UI navigation and answering complex queries with exact document citations, virtually eliminating hallucinations.<br>• Built a generative AI engine to dynamically synthesize customized, spaced-repetition flashcards and practice quizzes directly from user-uploaded lecture notes. | <b>Jan 2026</b> |
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**Distributed Healthcare Network | C++, TCP/UDP, SHA-256 — Systems & Security**

**Jan 2026**

- Architected a multi-tier distributed network comprising a client UI, central router, and three scalable backend services to optimize high-throughput data routing.
- Managed concurrent TCP client connections and routed backend queries via UDP sockets, securing sensitive PII/PHI with SHA-256 cryptographic hashing for HIPAA compliance.
- Programmed low-level Unix socket routines and custom signal handlers to automate child process reaping, preventing memory leaks and ensuring fault-tolerant socket teardowns.

**DevPortalX | React, Spring Boot, OAuth 2.0, DynamoDB — Full-Stack & Security**

**Oct 2024**

- Engineered an enterprise developer portal and centralized Identity Provider (IdP) using React and Spring Boot to manage third-party API integrations.
- Architected a custom OAuth 2.0 authorization server utilizing Spring Security to enable Single Sign-On (SSO) for external client applications.
- Implemented role-based access controls (RBAC) by issuing and validating JWTs to secure API endpoints and user data.
- Integrated DynamoDB for highly scalable data storage alongside analytics tooling, providing end-users with actionable API usage metrics and streamlined API key provisioning.

**LEADERSHIP**

- **National Society of Black Engineers, Graduate Student Liaison USC Chapter**

**Apr 2025 - Apr 2026**

Bridged the undergraduate-graduate gap by organizing joint social initiatives, driving a 25% increase in graduate student event attendance.